

KOPAONIK, Serbia

WMO: 133780

Lat: 43.2841N

Lon: 20.8017E

Elev: 5620

StdP: 11.95

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.1	6.8	-4.4	5.4	11.2	0.7	7.0	14.0	25.4	25.4	23.4	26.6	7.3	60	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.6	72.7	57.4	69.8	56.7	67.3	56.1	60.8	66.8	59.2	66.0	57.8	64.6	7.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.7	91.0	63.3	56.8	85.1	61.8	55.2	80.1	60.2	29.8	67.0	28.6	65.9	27.5	64.9	67.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.7	18.7	16.3	DB	-2.1	76.4	4.5	3.4	-5.3	78.9	-7.9	80.9	-10.4	82.8	-13.7	85.3	(4)
(5)			WB	-2.7	63.5	4.3	2.3	-5.7	65.2	-8.3	66.5	-10.7	67.9	-13.8	69.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	23.6	24.9	29.1	37.5	45.5	52.2	56.0	56.6	48.8	42.0	34.4	26.5		
	DBStd	13.98	8.80	8.71	8.30	7.62	7.07	6.86	5.92	6.16	7.23	8.16	8.76	8.60		
	HDD50	4374	819	703	649	380	173	54	15	15	104	265	470	727		
	HDD65	9198	1284	1123	1114	826	603	384	284	269	486	715	918	1192		
	CDD50	665	0	0	0	4	34	122	201	218	69	15	2	0		
	CDD65	14	0	0	0	0	0	1	5	7	1	0	0	0		
	CDH74	42	0	0	0	0	0	1	15	24	2	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	7.4	7.7	8.5	8.5	8.1	7.3	6.4	6.3	6.2	6.7	7.1	8.3	7.8		
	PrecAvg	34.5	2.2	2.2	2.8	3.2	4.0	3.5	3.1	2.4	3.2	2.7	2.6	2.6		
(15) Precipitation	PrecMax	49.7	4.8	3.6	5.6	8.0	7.9	7.4	7.1	5.9	7.8	6.5	5.2	4.8		
	PrecMin	22.3	0.6	0.7	0.7	1.7	1.7	1.2	0.6	0.0	0.7	0.1	0.1	0.2		
	PrecStd	6.3	1.3	0.8	1.3	1.4	1.6	1.5	1.7	1.7	1.8	1.4	1.4	1.1		
	PrecStd	6.3	1.3	0.8	1.3	1.4	1.6	1.5	1.7	1.7	1.8	1.4	1.4	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.8	47.7	51.5	60.9	66.8	72.6	76.2	77.3	70.9	63.6	58.8	49.5		
		MCWB	34.6	37.3	40.5	46.9	53.2	58.5	57.9	57.2	55.7	48.3	44.7	36.0		
	2%	DB	40.4	42.8	46.5	55.6	62.1	69.1	72.3	73.1	66.4	60.0	52.5	43.0		
		MCWB	33.3	34.4	37.2	44.6	51.5	57.3	57.4	57.3	53.6	48.6	42.0	36.7		
	5%	DB	37.2	39.1	43.4	51.3	59.3	66.4	69.3	70.0	62.5	56.5	48.3	39.8		
		MCWB	33.1	33.0	36.3	42.2	50.0	56.5	57.0	56.4	52.4	47.2	39.9	34.8		
	10%	DB	34.8	35.8	40.4	47.9	56.7	63.7	66.6	67.0	59.4	52.9	45.1	37.1		
		MCWB	32.3	32.4	35.2	40.7	49.3	55.4	56.4	55.8	51.2	46.2	39.8	34.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.7	40.0	42.2	48.5	56.1	62.6	63.2	62.5	58.1	52.9	48.2	42.4		
		MCDB	40.8	44.8	47.7	56.5	61.8	68.9	69.1	67.8	66.8	57.8	54.6	43.5		
	2%	WB	35.8	36.8	39.4	45.8	53.6	59.9	60.8	60.3	55.7	50.8	45.1	38.6		
		MCDB	37.7	40.1	43.8	52.8	59.4	65.6	66.7	67.1	61.9	55.8	48.3	40.2		
	5%	WB	34.2	34.6	37.8	43.8	51.8	58.0	59.2	58.8	54.0	49.2	43.0	36.5		
		MCDB	36.0	37.2	41.5	49.6	57.0	64.1	65.9	66.0	59.7	53.5	45.5	38.2		
	10%	WB	32.4	32.9	35.9	41.8	50.1	56.2	57.6	57.4	52.6	47.6	41.1	34.7		
		MCDB	34.2	35.2	39.4	46.3	55.2	62.2	64.3	64.6	57.7	51.6	43.3	36.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	9.9	10.5	10.8	11.7	12.5	13.4	13.6	11.4	11.5	9.9	9.4		
		MCDBR	12.9	13.4	14.0	16.1	15.7	16.3	17.4	17.2	16.0	16.2	15.1	13.5		
		MCWBR	8.5	8.3	8.1	8.4	9.0	9.2	9.0	8.9	8.2	9.4	9.6	8.8		
		MCDBR	10.1	10.6	12.0	14.5	14.0	14.8	15.2	15.0	13.3	13.3	10.3	9.5		
	5% WB	MCWBR	7.7	7.7	8.1	8.7	9.2	9.7	9.6	9.5	8.5	8.9	7.8	7.6		
		MCWBR	7.7	7.7	8.1	8.7	9.2	9.7	9.6	9.5	8.5	8.9	7.8	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.266	0.286	0.316	0.357	0.353	0.346	0.346	0.354	0.338	0.311	0.285	0.265		
	taud		2.464	2.407	2.401	2.310	2.353	2.395	2.384	2.358	2.403	2.492	2.516	2.490		
	Ebn at Noon		281	292	294	288	291	293	291	285	282	279	271	271		
	Edh at Noon		23	29	33	39	38	37	37	37	32	26	22	20		
(44) All-Sky Solar Radiation	RadAvg		459	720	1072	1384	1689	1922	1986	1801	1262	880	558	397		
	RadStd		69	89	134	171	164	183	153	164	123	115	64	64		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.21	N/A	N/A	N/A	+1.06	+1.57	N/A	N/A	N/A	N/A
(47)	Regional (41 neighbors)	+1.08	N/A	N/A	N/A	+0.63	N/A	N/A	-216	N/A	N/A

Nomenclature: See separate page